

DOCKHUB · REQUIREMENTS

DockHub requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3. R6 relaxed to 10 % and R16 redefined to the two-bay fit (DKH-DDR-001); R2 to SwapCell interface v0.3; status from DKH-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

DockHub requirements

These requirements are checked by calculation in DKH-CAL-001 (TRL 3). They are not yet validated with riders, hosts or a city and will be revised after co-design sessions (see DKH-PRB-001). On 2026-09-25 Amish accepted the review recommendations (DKH-DDR-001, DKH-DDR-002): R6 is relaxed from 20 % to 10 %, R16 covers the four-bay cabinet with two bays fitted, and R2 cites SwapCell interface v0.3 (DKH-DDR-001 item 12). These are decided. Every other target remains as proposed at TRL 2 and will be revised after co-design.

On paper, 9 of 16 requirements are met. **R6 and R9 are not met**; R8, R15 and R16 are at risk; R7 and R14 cannot be verified until hardware exists.

The **design duty case** used throughout is 20 swaps a day, each returning a SwapCell pack at about 15 % state of charge, in a street location with 4.5 peak sun hours and ambient air between -10 and 45 °C. Because a swap needs an empty bay, a station with n bays holds $n - 1$ packs.

Table 1. Requirements. Status is from DKH-CAL-001.

ID	Requirement	Target	Verification	Status at TRL 3 (DKH-CAL-001)
R1	Fast swap	60 s or less from tapping the reader to closing the second door, for a rider familiar with the station	Task analysis; later timed trials with riders	Met (31 s)
R2	SwapCell compatibility	Every bay builds to SwapCell interface v0.3: envelope, guide faces, blind-mate receptacle with the 10 k Ω INTERLOCK coding resistor, class D catch, CAN heartbeat and message set; no change to the interface	Interface review with the SwapCell project	Met by design review; one CAN channel per bay
R3	Charge speed	20 to 80 % in 1.5 h or less; full charge in 2.5 h or less, per bay, at 25 °C	Charge profile calculation	Met (1.2 h and 2.3 h)
R4	Throughput	20 or more swaps a day from 4 bays at the design duty case	Day model with a demand profile	Met by the four-bay fit (20 of 20 served, longest wait 4 min; ceiling 36 a day). The two-bay first prototype serves 9 of 20
R5	Grid supply	One single-phase branch circuit (120 V 15 A or 230 V 10 A); peak draw 1.5 kW or less	Load calculation	Met (1.25 kW; 11.0 A at 120 V) with chargers of power factor 0.9 or better
R6	Solar contribution	Canopy solar supplies 10 % or more of daily charging energy at the design duty case (relaxed from 20 %, DKH-DDR-001)	Day model	Not met (9.3 %; 11.4 % if flat packs are routed to bay 1 in daylight, which turns one rider in 20 away)
R7	Fire detection and response	Detect a pack or bay over 60 °C, or smoke, within 10 s; open that bay's charge contactor within 1 s of detection; sound a local alarm and send a remote alert; vent gas to the rear roof louver, away from the user side	Latency budget; later bench test with a heat source	Not verifiable at TRL 3 (electronic latency 3.1 s, contactor 0.15 s)
R8	Fire containment	A thermal runaway in one bay does not spread to a neighboring bay or pack for 30 min or more, and flame does not leave the cabinet front	Later propagation test by an accredited lab	At risk ; not verifiable at TRL 3
R9	Climate	Enclosure IP54 or better; operate at -10 to 45 °C ambient; never charge a pack outside the pack's reported charge temperature window	Thermal calculation; later chamber test	Not met : charging pauses above about 29 to 39 °C ambient (pack charge limit 45 °C) and for cold-soaked packs below about -1 °C
R10	Security	1.5 mm or thicker steel; fail-secure bay locks; cabinet anchored to its pad with tamper-resistant fixings; a pack is released only against a valid token and a returned pack	Design review	Met by design review
R11	Accessibility and use	All operable parts (reader, buttons, bay handles) 0.38 to 1.22 m (15 to 48 in) above	Model check; later user trials	Met (0.73 to 1.10 m)

ID	Requirement	Target	Verification	Status at TRL 3 (DKH-CAL-001)
		the ground; status shown by light and symbol as well as text; usable with gloves and at night		
R12	Privacy and data	No cameras or microphones; logs hold pack ID, bay, time, energy and state of health, with the rider as an opaque token; operates offline for 24 h or more and syncs later	Design review	Met by design review
R13	Sidewalk footprint	Cabinet footprint 1.0 x 0.6 m or less; canopy underside 2.1 m or more above the sidewalk	Model check	Met (1.0 x 0.5 m with a flush plinth; lowest canopy point over the sidewalk 2.18 m)
R14	Maintainability	Any charger, lock, bay cradle or controller replaced in 15 min or less by one technician with hand tools, through the service door	Design review; later timed trial	Not verifiable at TRL 3
R15	Wind and anchoring	Stays upright and intact at a 30 m/s gust with the canopy fitted	Wind load calculation	At risk: overturning 2.9 kN·m against 0.50 kN·m self-weight; needs four anchors of 3.8 kN design tension and a pad, not yet chosen
R16	Cost	Parts 1,200 USD or less for the four-bay cabinet with two bays fitted (first prototype), SwapCell packs excluded (redefined, DKH-DDR-001)	Priced BOM (bom/bom.csv)	At risk (\$1,199; \$1,423 with all four bays fitted)

■ Assumptions

- SwapCell pack (interface v0.3): 46.8 V nominal, about 468 Wh, 2.85 kg, charge on a certified 54.6 V 5 A charger in 2.3 h to full and 1.2 h for 20 to 80 %, charge window 0 to 45 °C (SWC-PRC-001 v0.3, SWC-CAL-001).
- Charger efficiency about 90 %, cell charge efficiency about 95 %, cabinet auxiliary load about 30 W on average and 40 W peak.
- 400 W panel, 4.5 peak sun hours, 0.8 derate, 96 % MPPT, feeding bay 1 only.
- Packs are released at 80 % state of charge or more, fullest first; riders arrive with a lunch and a dinner peak and leave after waiting 15 min (DKH-CAL-001 Table 1).
- Delivery riders cover roughly 60 to 100 km a day at about 10 Wh/km. This is an assumption to check with riders; it implies 1 to 2 swaps per rider per shift.
- Accessibility reach range from the US ADA forward reach limits (15 to 48 in); local rules may differ.